

Static Cosmology

Lecture 1: Observational Coordinates in a Static Universe

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Overview

A static, spatially infinite model is written in observational coordinates with a time-independent mean density.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$\partial_t \langle \rho_m \rangle = 0$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$dl^2 = \delta_{ij} dx^i dx^j$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$I_\nu(s) = I_\nu(0) e^{-\int \kappa_\nu \rho_m ds}$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 1
- CMT-9 intro
- Catalog Calibration Handbook §2